

Jackson Ferguson

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Education

University of Victoria

Victoria, BC

B.Sc. Major Combined Physics and Astronomy

Graduated Spring 2026

High Performance Coursework: Quantum Computing (100%), Stellar Astrophysics (99%), Probability & Statistics (97%), Computer Assisted Physics (91%), Experimental Physics (88%), Thermodynamics (86%).

Technical Skills

Hardware & Instrumentation: RP2040/ARM Architecture, Oscilloscopes, DAQ, Soldering, BOM Management
Languages & Firmware: Python, MicroPython, Bash, Serial Comm (UART), $\LaTeX$
Signal Processing (DSP): SciPy, NumPy, Time-domain analysis, FFT, Custom Windowing
Toolchains & Automation: Git/GitHub Actions, Docker, CLI Tooling, Subprocess Management, uv

Engineering & Research Projects

Systems Audio Lab: Precision DAQ & Analog Front End

Mixed-Signal Design, Instrumentation

- **Embedded Instrumentation:** Engineered a custom high-speed data acquisition system (DAQ) using an RP2040; implemented a zero-jitter, store-and-forward memory architecture to decouple hardware sampling from host OS scheduling.
- **Analog Front End (AFE):** Designed and fabricated a conditioning circuit with switchable impedance and AC/DC coupling; utilized silicon diode clamping ($I_R < 25\text{ nA}$) over Schottky to eliminate DC offset leakage in high-Z paths.
- **Infrastructure & Thermal Management:** Designed a custom 12 V-to-9 V linear voltage regulator with calculated thermal dissipation limits to eliminate switching ripple from sensitive analog measurements.
- **Empirical Validation:** Verified system precision by characterizing a CMOS saturation circuit; successfully isolated a 1.3 mV RMS noise floor to mathematically validate even-order harmonic distortion signatures.

Star Ground: Physical-Digital Isomorphism Engine

Python, Hardware Integration, Domain Logic

- **Z-Height Sorting:** Architected a physical-digital translation layer that parses unstructured BOMs and reorders assembly instructions based on physical component Z-height, optimizing operator workflow during PCB population.
- **Domain-Aware Heuristics:** Developed a classification system that infers analog circuit topologies from raw text, auto-injecting necessary enclosure hardware and recommending audio-grade silicon substitutions.
- **Deterministic Hardware Logistics:** Replaced error-prone manual part aggregation with a strictly validated dependency tree, actively mitigating the risk of critical assembly-halting stockouts during physical prototyping.

Protostar: Toolchain & Environment Scaffolder

Python, State-Machine Logic

- **Toolchain Automation:** Architected a deterministic bootstrapping tool to automatically configure isolated host environments for MicroPython/RP2040 firmware development and DSP data pipelines.
- **State-Machine Execution:** Designed a fault-tolerant orchestrator that strictly decouples state declaration from disk I/O, enforcing dependency sequencing and pre-flight system validation before execution.
- **Data Integrity:** Built a robust parsing engine capable of safely merging host configurations, preventing data corruption or systemic overwrites when interfacing with dynamic, pre-existing workspace constraints.

Git Pulsar: Hardware-Aware Background Daemon

Python, systemd, Hardware Telemetry

- **OS-Level Daemonization:** Developed a cross-platform background service integrated directly with host init systems (Linux systemd user timers, macOS launchd) for persistent, out-of-band state capture.
- **Hardware Telemetry Polling:** Built dynamic, power-aware execution loops that poll hardware state (parsing Linux `/sys/class/power_supply` and macOS `pmset`) to throttle network I/O and suspend operations during low-battery states.

Galaxy Cluster Dark Matter Analysis (ACO 2670)

Astrophysics Research Project

- **Pipeline Development:** Constructed a data pipeline to clean spectroscopic data for 98+ galaxies, utilizing redshift-space distortion analysis to isolate cluster members from the background field.
- **Statistical Modeling:** Applied the Virial Theorem and Gaussian Mixture Models (GMM) to calculate a mass-to-light ratio of $(291 \pm 60) M_{\odot}/L_{\odot}$, providing kinematic evidence of Dark Matter dominance.

Work & Leadership Experience

Co-Lead Event Organizer, Unplugged Events, Victoria, BC

Jan 2025 – Present

- Managed technical production, signal flow, and venue logistics for live events (100+ attendees).

Server, Stanley Park Brewing, Vancouver, BC

May 2025 – August 2025

Alpine Ski Instructor, Whistler Blackcomb, Whistler, BC

Nov 2018 – Feb 2020

Freelance 3D Modeler, Twisted Time Studios, Remote

May 2015 – Aug 2015

- Designed and created main character models and weapon systems for Extinction: Archaic Genesis.